

Synoptic CB: Macrophyte Fluxes

2022 Samples

2026-02-23

Contents

0.1	Write out files	2
0.2	All Data by Year, Site and Zone	2
0.3	Boxplot summary by Date	3
0.4	Boxplot summary by Site	4
0.5	Summary Tables	5

#Read in and collate all the files in the Processed Data folder

```
ch4 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2022_CH4.csv")
ch4$X <- NULL
ch4 <- ch4 %>%
  rename_with(
    .cols = 7:(ncol(.) - 3),
    .fn = ~paste0("ch4_", .)
  )
```

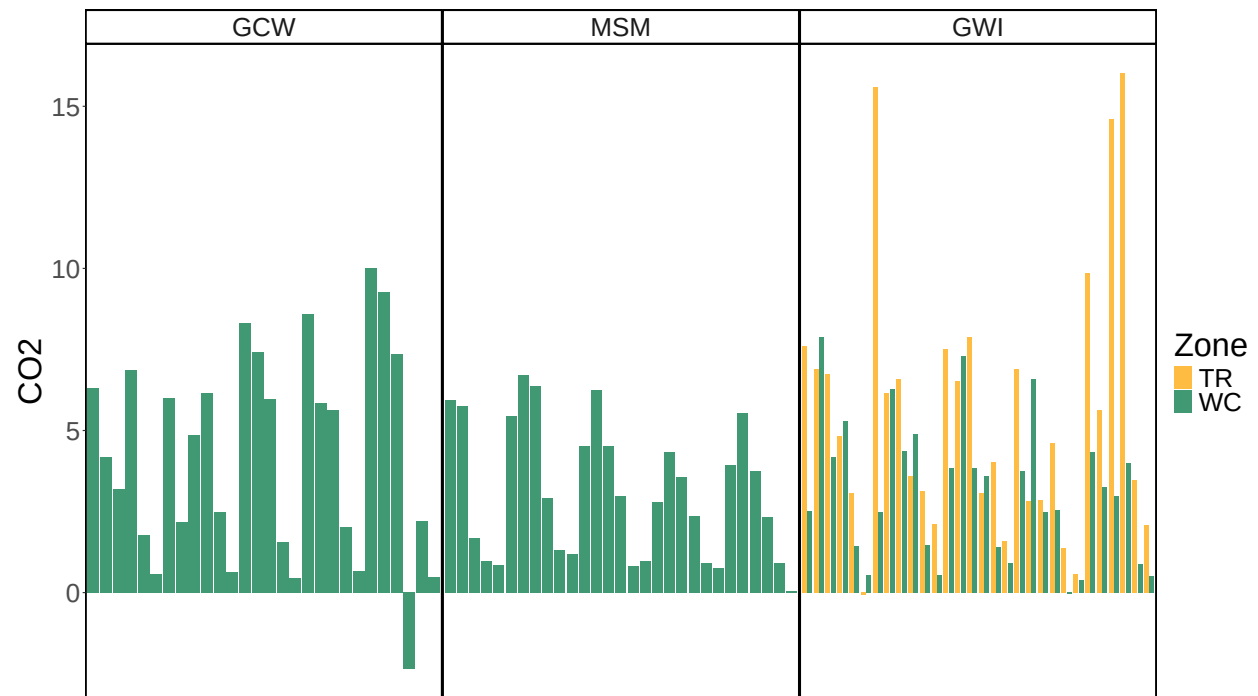
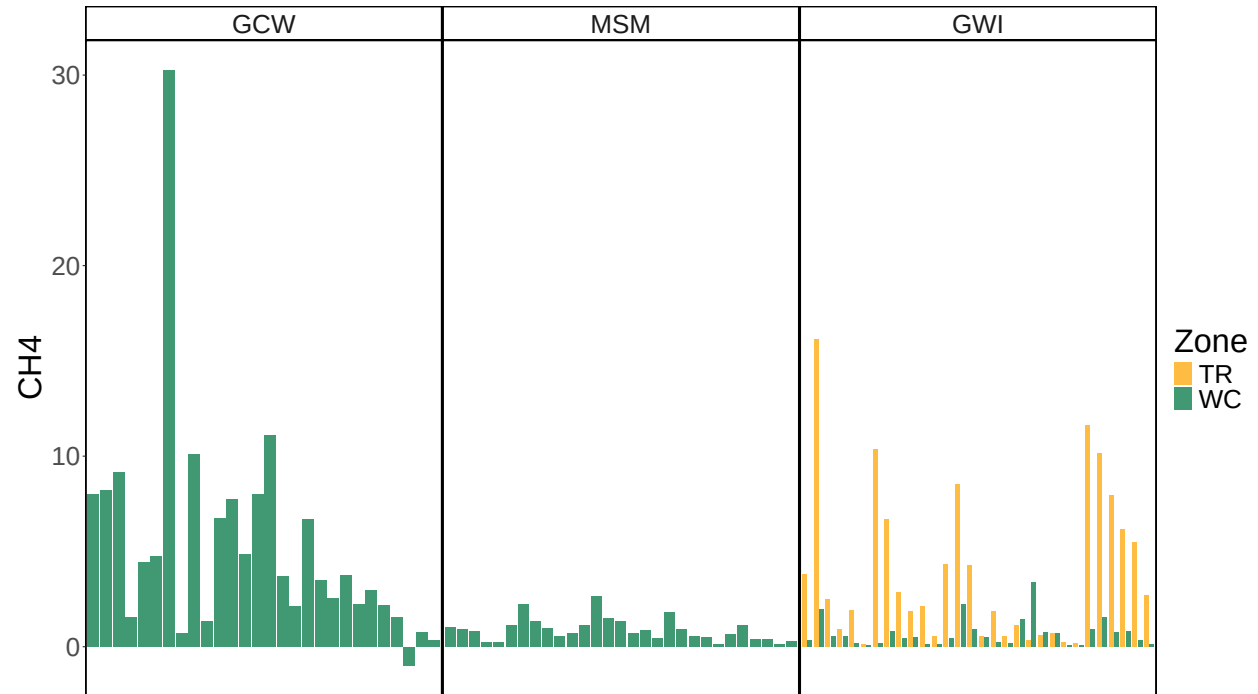
```
co2 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2022_CO2.csv")
co2$X <- NULL
co2 <- co2 %>%
  rename_with(
    .cols = 7:(ncol(.) - 3),
    .fn = ~paste0("co2_", .)
  )
```

Combine all files

```
all_dat <- merge(ch4, co2, by = c("Plot", "Site", "Zone", "Replicate", "Date", "TIMESTAMP", "TIMESTAMP_1"))
```

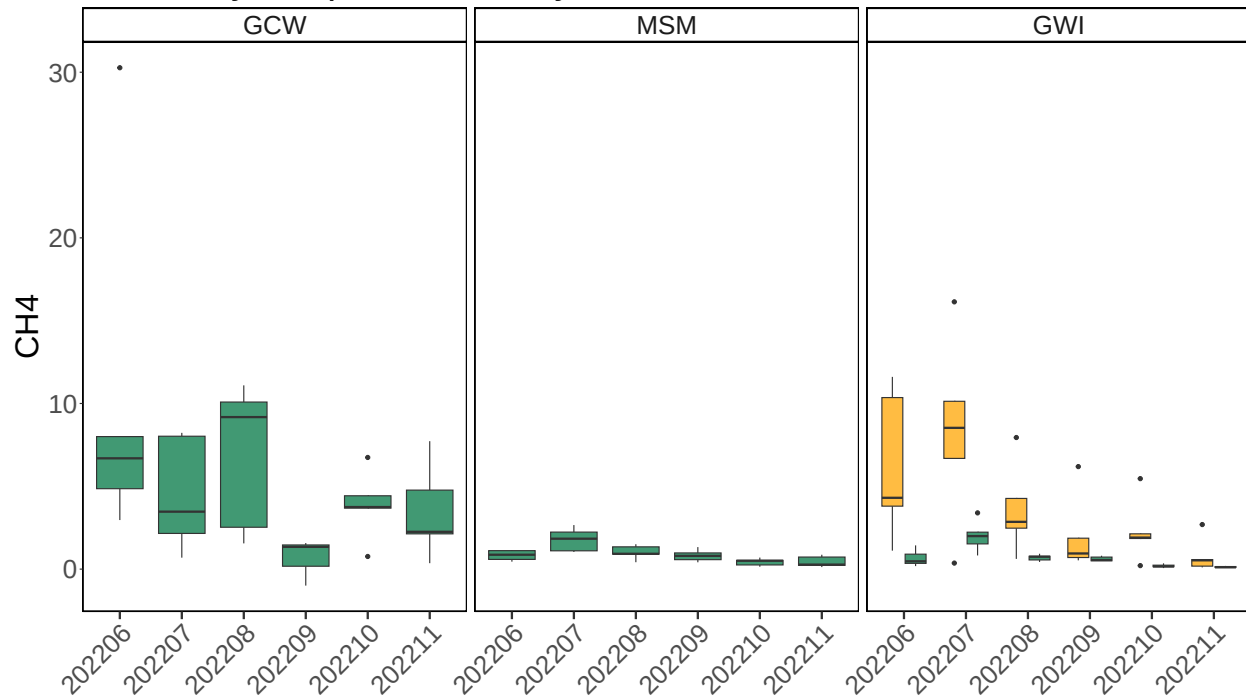
0.1 Write out files

0.2 All Data by Year, Site and Zone

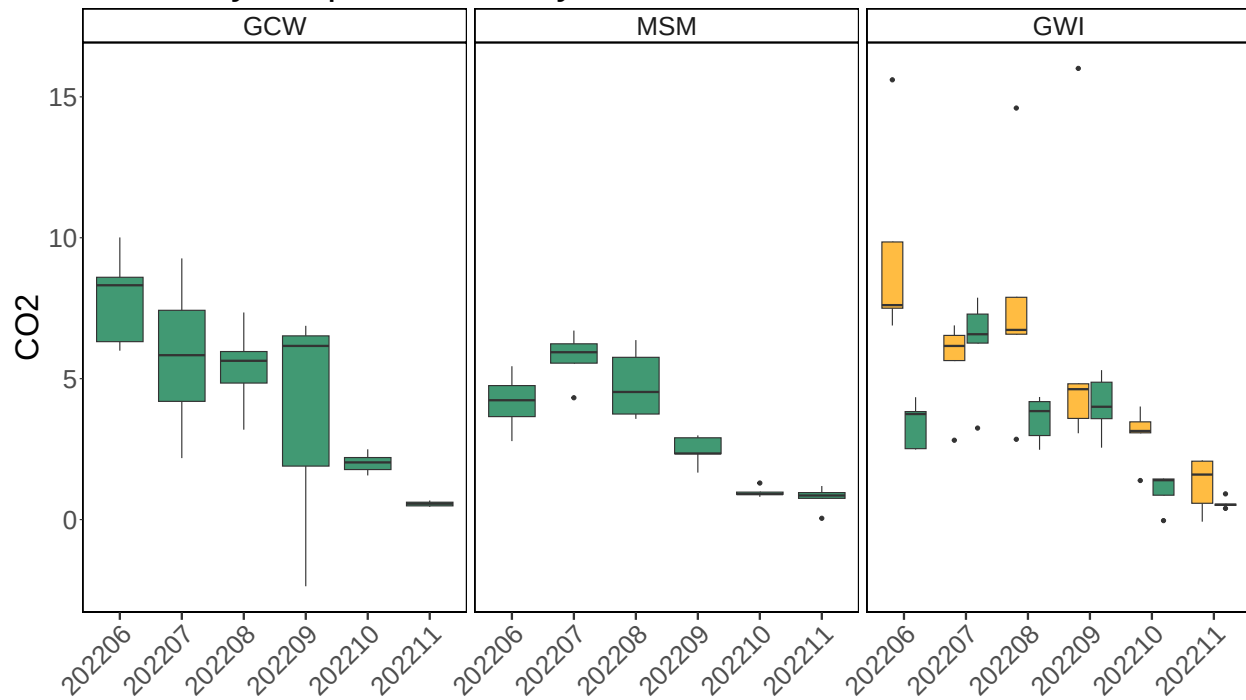


0.3 Boxplot summary by Date

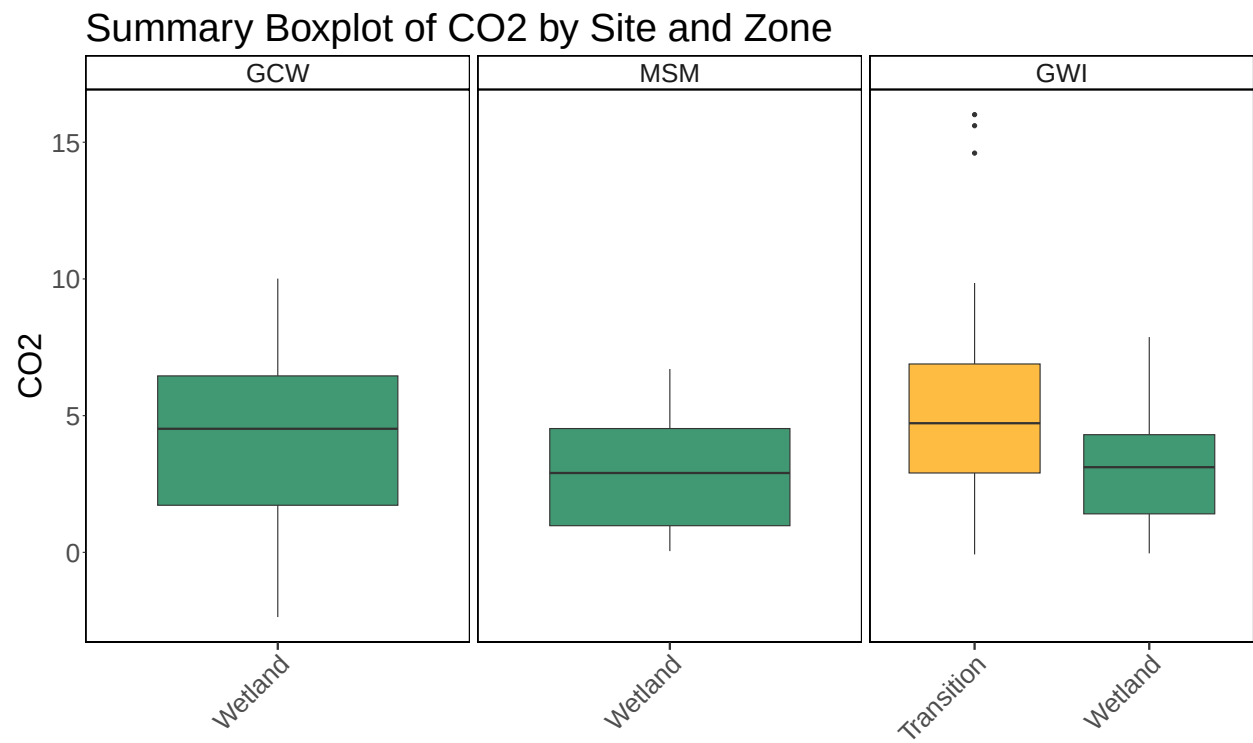
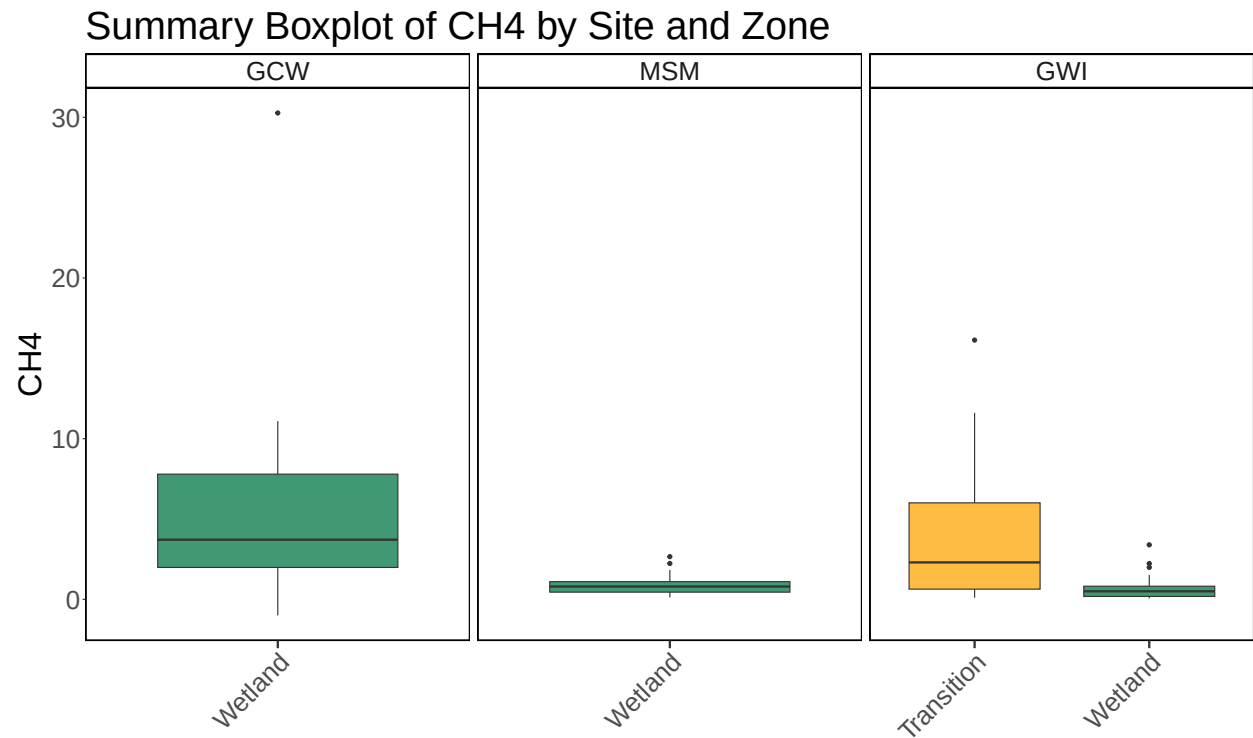
Summary Boxplot of CH4 by Site and Zone



Summary Boxplot of CO2 by Site and Zone



0.4 Boxplot summary by Site



##Calculate summary metrics

0.5 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
GCW	Wetland	-0.9955924	30.270376	5.2966172	-2.3658928	10.011470	4.237311
MSM	Wetland	0.1235615	2.659060	0.8872488	0.0458841	6.706775	3.115307
GWI	Transition	0.1053134	16.136358	3.9018021	-0.0745288	16.009112	5.587092
GWI	Wetland	0.0628300	3.394554	0.7146406	-0.0364413	7.873652	3.144701